

[Aim: 100/100 in Maths] ✓

आरंभ

REAL NUMBERS

Lecture #2

#ATK2:-

→ LCM $\Rightarrow$ वो सबसे छोटा No. जो दोनों से divisible हो।
HCF $\Rightarrow$ वो सबसे बड़ा No. जो दोनों को divide करे।

$$\begin{array}{r|l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$
$$\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 2 & 3 \\ \hline & 1 \end{array}$$

Short Trick

$$8 = 2 \times 2 \times 2 \Rightarrow 2^3$$

$$12 = 2 \times 2 \times 3 \Rightarrow 2^2 \cdot 3^1$$

$$\underline{HCF(8, 12)} = 2 \times 2 \Rightarrow 4$$

$$\underline{LCM(8, 12)} = 2^3 \cdot 3^1 \Rightarrow 8 \times 3 \Rightarrow 24$$

#Prime Numbers: having only two factors $\swarrow \begin{matrix} ① \\ \text{itself} \end{matrix}$



Prime no $\rightarrow$

2, 3, 5, 7, 11, 13, 17, ...

#Composite Numbers: having more than two factors.

$\rightarrow$ 4, 6, 8, 9, 10, 12, 14, 15, ...

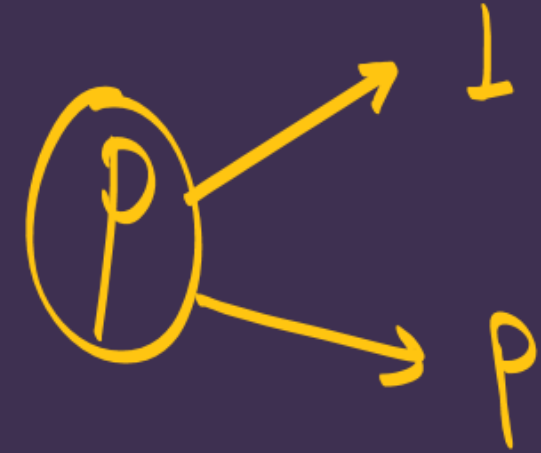
 $\rightarrow$ 1 is neither prime nor composite.

#LP: The natural number 1 is : [CBSE 2026]

- (A) a prime number.**
- (B) a composite number.**
- (C) prime as well as composite.**
- (D) ✓ neither prime nor composite.**

#LP: Let p be a prime number, then sum of its factors is :

- (A) p
- (B) 1
- ☒ (C) $p+1$
- (D) $p-1$



$$\boxed{\text{Sum} = 1 + p}$$

#LP: The ratio of HCF to LCM of the least composite number and the least prime number is: [CBSE 2023]

(A) 1:2

(B) 2:1

(C) 1:1

(D) 1:3

$$\frac{\text{HCF}(4, 2)}{\text{LCM}(4, 2)} = \frac{2}{4} \Rightarrow \frac{1}{2}$$

1:2

$$\begin{array}{r|l} 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 2 \\ \hline & 1 \end{array}$$

$$4 = 2 \times 2 \Rightarrow 2^2$$

$$2 = 2$$

$$\text{HCF} = 2$$

$$\text{LCM} = 2^2 \Rightarrow 4$$

#LP: If the HCF of 408 and 1032 is expressible in the form $1032m - 408 \times 5$, find m .

$$\begin{array}{r} 408 \\ 204 \overline{) 408} \\ \underline{204} \\ 204 \end{array}$$

prime factorisation

$$1032 = 2 \times 2 \times 2 \times 3 \times 43$$
$$408 = 2 \times 2 \times 2 \times 3 \times 17$$

prime factorisation

$$\text{HCF} = 2 \times 2 \times 2 \times 3$$

$$\begin{array}{r} 2 \overline{) 408} \\ \underline{204} \\ 204 \\ \underline{204} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 1032} \\ \underline{206} \\ 826 \\ \underline{826} \\ 0 \end{array}$$

ATQ $\Rightarrow 1032m - 408 \times 5 = 24$

$$1032m - 2040 = 24$$

$$1032m = 24 + 2040$$

$$1032m = 2064$$
$$m = \frac{2064}{1032} = 2$$

$m = 2$

Prime factorisation

prime
no. को $\rightarrow$ multiply में लिखना

#LP: If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; x, y are prime numbers, then HCF (a, b) is [NCERT Exemplar]

(A) xy

(B) xy^2 ✓

(C) x^3y^3

(D) x^2y^2

$$\begin{aligned} a &= x^3 y^2 \Rightarrow \underbrace{x \cdot x \cdot x}_{\text{prime fact.}} \cdot \underbrace{y \cdot y}_{\text{prime fact.}} \\ b &= x y^3 \Rightarrow x \cdot \underbrace{y \cdot y \cdot y}_{\text{prime fact.}} \end{aligned}$$

$$\begin{aligned} \text{HCF} &= x \cdot y \cdot y \\ &= x \cdot y^2 \end{aligned}$$

#LP: Let a and b be two positive integers such that $a = p^3 q^4$ and $b = p^2 q^3$, where p and q are prime numbers. If $\text{HCF}(a, b) = p^m q^n$ and $\text{LCM}(a, b) = p^r q^s$, then $(m+n)(r+s) =$

(A) 15

(B) 30

(C) 35 ✓

(D) 72

$$a = p^3 q^4 \Rightarrow p \cdot p \cdot p \cdot q \cdot q \cdot q \cdot q$$

$$b = p^2 q^3 \Rightarrow p \cdot p \cdot q \cdot q \cdot q$$

$$\text{HCF} = p^2 \cdot q^3 = p^m q^n$$

$$\text{LCM} = p^3 \cdot q^4 = p^r q^s$$

$$\begin{aligned} m &= 2 \\ n &= 3 \\ r &= 3 \\ s &= 4 \end{aligned}$$

$$(m+n)(r+s)$$

$$(2+3)(3+4)$$

$$5 \times 7$$

$$\Rightarrow \underline{\underline{35}}$$

#LP: If H.C.F. of $135x^2$ and $189x^3$ is 108, then find the value of x.

[CBSE 2026]

$$\begin{aligned} 135x^2 &= 3 \times 3 \times 3 \times 5 \cdot x \cdot x \cdot x \\ 189x^3 &= 3 \times 3 \times 3 \times 7 \cdot x \cdot x \cdot x \cdot x \end{aligned}$$

$$\text{HCF} = 3 \cdot 3 \cdot 3 \cdot x \cdot x$$

$$108 = 27x^2$$

$$4 = x^2$$

$$x = \sqrt{4} = \pm 2$$

$$\begin{array}{r|l} 3 & 189 \\ \hline 3 & 63 \\ \hline 3 & 21 \\ \hline 7 & 7 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 3 & 135 \\ \hline 3 & 45 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

#LP: The least number that is divisible by all the numbers from 1 to 10 (both inclusive) is

(A) 10

(B) 100

(C) 504

(D) 2520 ✓

LCM

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$$
$$=$$

2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
2	1, 1, 3, 2, 5, 3, 7, 4, 9, 5
2	1, 1, 3, 1, 5, 3, 7, 2, 9, 5
3	1, 1, 3, 1, 5, 3, 7, 1, 9, 5
3	1, 1, 1, 1, 5, 1, 7, 1, 3, 5
5	1, 1, 1, 1, 5, 1, 7, 1, 1, 5
7	1, 1, 1, 1, 1, 1, 7, 1, 1, 1

$$\begin{array}{r|l}
 2 & 6, 8 \\
 \hline
 2 & 3, 4 \\
 \hline
 2 & 3, 2 \\
 \hline
 3 & 3, 1 \\
 \hline
 & 1, 1
 \end{array}$$

$$\begin{aligned}
 L(m) &= 2^4 \times 2 \times 2 \times 3 \\
 &\rightarrow \textcircled{24}
 \end{aligned}$$

Part - 1 (Basic Multiple Choice Questions)

3. If the HCF of 567 and 693 is expressible in the form $567x + 693 \times (-4)$, find x.

(a) 4

(b) 2

~~(c) 5~~

(d) 3

6. The sum of exponents of prime factors in the prime factorisation of 196 is

(a) 3

~~(b) 4~~

(c) 5

(d) 2

$$196 = 2^2 \times 7^2$$

$$2 + 2 \Rightarrow 4$$

2	196
2	98
7	14
7	2
	1

7. The LCM of three numbers 28, 44, 132 is :

(a) 258

(b) 231

(c) 462

(d) 924



Part - 2 (Basic Subjective Type Questions)

4. If $a = 2^3 \times 3$, $b = 2 \times 3 \times 5$, $c = 3^n \times 5$ and $\text{LCM}(a, b, c) = 2^3 \times 3^2 \times 5$, then n is equal to

$$a = 2^3 \times 3^1$$

$$b = 2^1 \times 3^1 \times 5^1$$

$$c = 3^n \times 5^1$$

$$\text{LCM} = 2^3 \cdot 3^n \cdot 5^1$$

$$= 2^3 \cdot 3^2 \cdot 5^1$$

on comparing $\rightarrow n = 2$

Part - 3 (Advanced Miscellaneous Questions)

7. Match the Column I with Column II and choose the correct option :

Column-I	Column-II
(A) The HCF of 12, 21 and 15 is	(i) $3^2 \times 5^2$
(B) HCF of 112 and 198 is	(ii) 3
(C) 225 can be expressed as	(iii) 2
(D) 196 can be expressed as	(iv) $2^2 \times 7^2$

- (a) A-iv, B-ii, C-i, D-iii
- (b) A-i, B-iv, C-iii, D-ii
- (c) A-ii, B-iv, C-i, D-iii
- (d) A-ii, B-iii, C-i, D-iv

(Case Based Questions)

1. LCM of several fractions = $\frac{\text{LCM of their numerators}}{\text{HCF of their denominators}}$

$$\text{LCM} = \frac{\text{LCM}}{\text{HCF}}$$

HCF of several fractions = $\frac{\text{HCF of their numerators}}{\text{LCM of their denominators}}$

$$\text{HCF} = \frac{\text{HCF}}{\text{LCM}}$$

(i) The LCM of the fractions $\left(\frac{5}{6}, \frac{15}{24} \text{ and } \frac{25}{8}\right)$ is $\Rightarrow \frac{\text{LCM}(5, 15, 25)}{\text{HCF}(6, 24, 8)} \Rightarrow$

(a) $\frac{5}{48}$

(b) $\frac{5}{8}$

(c) $\frac{75}{48}$

(d) $\frac{75}{8}$

(ii) The HCF of $\left(\frac{2}{5}, \frac{6}{25} \text{ and } \frac{8}{35}\right)$ is $\Rightarrow \frac{\text{HCF}(2, 6, 8)}{\text{LCM}(5, 25, 35)} \Rightarrow$

(a) $\frac{2}{5}$

(b) $\frac{24}{5}$

(c) $\frac{2}{175}$

(d) $\frac{24}{175}$

(iii) The HCF of the fractions $\frac{8}{21}, \frac{12}{35}$ and $\frac{32}{7}$ is

(a) $\frac{4}{105}$

(b) $\frac{192}{7}$

(c) $\frac{3}{105}$

(d) $\frac{5}{109}$

#LP 5: The LCM of 20, 28 and 36 is:

#LP 6: If the HCF of 408 and 1032 is expressible in the form:
 $1032m - 408 \times 5$, find m .

#LP 7: If two positive integers p and q are written as $p=ab^2$ and $q=a^3$
 b, a, b are prime numbers, then LCM (p, q) is

- (A) ab
- (B) $a^2 b^2$
- (C) $a^3 b^2$
- (D) $a^3 b^3$

#LP 8: If two positive integers a and b are written as $a = x^3 y^2$ and $b = xy^3$; x, y are prime numbers, then HCF (a, b) is

- (A) xy
- (B) xy^2
- (C) $x^3 y^3$
- (D) $x^2 y^2$